

# The Linux Way

Here's how you would create an image of a hard disk, or of any storage device for that matter, in Linux.

**Step 1:** The command `dd if=/dev/hda of=harddisk.img` copies the entire hard disk to a file called `harddisk.img`. You can similarly make an image of any device, including CDs, floppies or whatever you choose. Once you're through making the image, you can mount it through a loopback device, allowing you to edit the image file.

**Step 2:** Use the command line `mount -t ext2 -o loop harddisk.img /mnt/loop` to mount the image. After you mount it, you can remove all the files you don't really need restored, and thereby save space in your image file.

The best part though, is that you do not need third-party software to make a drive image in Linux—the feature comes inbuilt.

programs, etc. You can then easily restore your system to its original condition.

## Cloned applications to the rescue

PowerQuest's DriveImage Pro (PQDI Pro) is one of the many software that allows you to create a disk image (read clone) of a Windows partition and store it onto a different partition on the same hard drive. You can take a drive image of the partition once you've installed Windows, device drivers and Office applications, configured your e-mail, networking and Internet settings, and tweaked your machine. This process copies all the contents of the partition, including the boot settings, configuration and system files into a single file that can be compressed to 50 per cent of the actual size. This compression takes place simultaneously while the image is being created.

Before you start off with cloning your system, you need to ensure that you have completed installing and configuring your machine right from scratch. Once you have this out of the way, install PQDI's Drive Image Pro onto your hard drive. Just follow the on-screen prompts and keep pressing Next. You will also need a formatted 1.44 MB floppy disk—preferably more than one, just to be safe.

## Creating an image

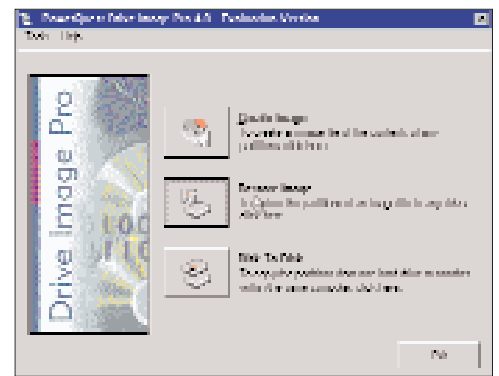
PQDI works best if run through MS-DOS mode and that too via a floppy disk as this allows it full access to the partition that you want to clone.

**Step 1:** From the PQDI Program group in the Windows Start Menu, choose the Boot Disk Builder option. You can either use a standard Win95/98/2000/Me boot disk or use PQDI to make one. It is recommended that you use PQDI to make the disk as it takes care of the entire process. If you were to use a standard boot disk, you would have to do it manually, that is, you would have to type your way to the folder where you installed PQDI and then finally execute PQDI.

**Step 2:** Under Boot Disk Builder, select the Stand-alone Boot Disks option, click Next and follow the on-screen messages to complete the process.

The resultant floppy disk contains the files that PQDI would need to make a drive image. Store this disk in a safe place, as this will be required when you decide to restore the image.

**Step 3:** Insert the disk in your floppy drive and restart your PC. The floppy disk will automatically start PQDI. Here, click on 'Create Image' and hit Next.



Creating an image using PQDI

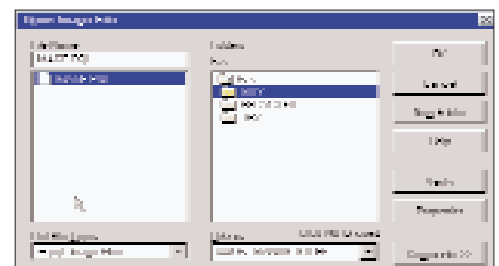
**Step 4:** Specify the drive/partition that you want to clone. You can also specify a password for the image file so that no one else can use that image. Click OK and then click Next to continue.



Specifying the drive to be cloned

**Step 5:** Specify the name and a path where you want to save the image and continue.

**Step 6:** Set the compression level to Low, High or No, depending upon the amount of free space you have. Bear in mind that an image made using 'High' compression will take the longest to restore. This is because during restoration, PQDI will first have to uncompress and then restore. In our case we chose High.



Setting the compression level

**Step 7:** Finally, press Next twice and PQDI will start creating and saving the image to the location you specified in Step 6.

Once the process is over, you will get a message confirming successful completion of the process



Creating a boot disk using PQDI